Multimedia, communication, and an experiential environment

Multimedia (多媒体): the combined use of several media that stimulate one or more of the five senses (sight, hearing, touch, smell, taste) to deliver relevant information to users. *e.g.* a dentist's "Just lie down, shut up, and open wide" needs the scene to make sense. Each sense gives only partial information. Not just "entertainment/music/photo/video/ebook" (narrow views). Audio/video dominate because humans analyse them well and they are easier to capture/reproduce than smell/taste/touch.

Computing (计算): any goal-oriented activity requiring, benefiting from, or creating computing machinery. *e.g.* a navigation app computing the fastest route.

Multimedia computing: utilise signals from multifarious sensors and present to users only the relevant information in the appropriate sensory modality (media). Goals: information sharing on experiences; knowledge extracting from experiences. *e.g.* a photo app turns photos' time + GPS metadata into a trip route on a map.

Communication (沟通/传播): key to share experiences and to create, maintain, sustain and propagate knowledge; across time (preserve/replay), space (remote) and addressee (as many people as possible). *e.g.* a recorded lecture crosses time; uploading it reaches many students anywhere.

Invariance table (不变性) (slide): speech -> none; writing -> time; paper -> time+space; print -> time+space+addressee; telegraph/telephone -> space; radio/TV -> space+addressee; movies/recording/digital media -> time+space+addressee (digital adds machine enhancement/processing); Internet -> time+space+(personalized) addressee.

Development: application -> technology -> device: numerical -> ALU -> computer (1960s IBM 1620 "CADET", no ALU, lookup tables); business -> alphanumeric/database -> workstation/server; documents/Web -> connectivity/CPU -> PC/laptop; social/photo -> mobile comms/AV/geolocation -> phone/tablet. CPU, storage, bandwidth, and algorithm progress made audio/video practical.

Eras: 1960s IBM 1620 (lookup tables, punch cards) -> pre-1990s text-only PCs -> early 1990s sound cards, 3-D games -> mid-1990s Win95, CD-ROM -> end-1990s every PC is "multimedia" -> today phones have more sensors than humans; 1G voice only -> 3G/4G/5G video.

Emerging applications (新兴应用): live spatiotemporal streams are the norm; a holistic picture of events beats silos of isolated data; insights independent of medium/source; only relevant information counts; exploration (browsing) replaces relational database querying. Two aspects: the evolving nature of information and the expected results from computing. *e.g.* live traffic map vs a static yearly report.

Amazon example: (1) orders HK-Taiwan = static count; (2) regions with rapid growth = trend over time; (3) hottest item now = live, constantly changing.

Multimedia vs Monomedia (单媒体): monomedia (computer vision, audio processing) analyses one medium; slide's 3 points: (1) multimedia is closely related to how humans experience the world; (2) partial information from multiple media sources is correlated and combined to get complete information about the environment; (3) multimedia computing and communication is fundamentally about combining information from multiple sources in the context of the problem being solved.

Formal definition: synchronised sensor streams S_1..S_n of K data types plus **metadata** (元数据) M_i (annotations, parameters, time, location) -> feature detectors -> **features** (特征) F_ij (j-th feature stream of S_i) -> integrate and process (**association/fusion** (关联/融合)) -> insights -> user feedback (query) as a new input stream. Example: photo metadata (time, GPS) -> insight = route of a trip on a map. Bluetooth/NFC are communications, not sensors.

Experiential environment (体验式环境): an environment where users directly use their senses to observe data and information of interest related to an event, and interact naturally with the data based on their particular set of interests in the context of that event. *e.g.* swiping through a photo map of a trip to explore where and when each picture was taken.

Why needed: current information environments work against human-machine synergy: humans are good at conceptual/perceptual analysis, computers at mathematical/logical analysis. Experiential systems are technically implemented as multimedia systems (integrate, process, output data from different sensory modalities).

3 properties: (1) direct and intuitive: holistic picture, no unfamiliar metaphors/commands (swipe to next photo, tap app icon, Kinect); (2) same query and presentation spaces, WYSIWYG (spreadsheet: change a value, see new relationships); (3) consider user state and **context** (自适应), promote perceptual analysis and exploration (video games: audio + video + touch).

Experience, perception, semantics, and systems

Objective of a multimedia system: process different types of data streams simultaneously as one correlated set representing information/knowledge for solving a problem. *e.g.* lecture-recording software keeps mic audio and PC screen as one synchronised set.

Challenge: discover correlations in the multimedia data; combine partial information from disparate sources into holistic information in a given context. *e.g.* COVID gate: thermal sensor + camera mask check -> one decision.

Experience (经验): direct observation of, or participation in, events as a basis of knowledge. *e.g.* living through a pandemic.

Information (信息): an efficient but abstract communication of experience. *e.g.* the sentence "wearing a mask is a powerful weapon".

Knowledge (知识): accumulated/aggregated experiences. *e.g.* a doctor's expertise built from many cases.

Communication process: sharing experiences and information with others (who may be separated in space and time); we do not communicate the experience itself, so the act of experiencing is lost (mask example); effective communication needs relevant experience.

DIKW: data records -> information gives meaningful relationships -> knowledge explains patterns -> wisdom chooses action under goals.

Dictionary (词典): a shared and agreed-upon collection of symbols (words) and what they mean: language = words; CS = codes/terms/keys, also dictionary compression; multimedia = **percepts** (感知元素) (visual dictionary: curves, lines, shapes).

Objects (对象) / events (事件): objects capture static components of the world; events represent changes in relationships among objects; audio/video are dynamic (attributes as functions of time). "Person" is an object; "person enters a restricted zone" is an event.

Perception (感知): the process of understanding sensory signals to recover useful information; signal + existing knowledge -> information. It generates **hypotheses** (假设) and selects the best supported; **controlled hallucination** (受控幻觉) = model-guided interpretation, not arbitrary invention. Without knowledge, no information.

Neisser cycle (奈瑟知觉循环): schema (图式) directs exploration -> exploration samples environment -> environmental information modifies schema -> revised exploration.

（已有认知指导观察，新的观察又修正认知。） The agent decides what extra information is needed and how to get it. Vase/faces, Dalmatian, and "Wine or Soy Sauce / MTR Mei Foo" show prior knowledge + context drive interpretation.

Semantic gap (语义鸿沟): the difference between the information a machine can extract from (perceptual) data and the interpretation a user in a given situation has for the same data. Machine: bits/bytes -> lists/images/audio/video; user: objects/events -> knowledge. Reduce it with content + context + metadata + priors + user feedback; it never fully disappears. Computers see only numbers (25 C), not "hot/cold", until we give thresholds (<20 cold, >28 hot). Transformers succeed because they model context.

Content (内容) / context (情境/上下文) / metadata: content = acquired data; context = interrelated conditions in which content is acquired; metadata = "data about data", providing context parameters (photo Exif time/location/camera). Combine content + context.

Five modules (五大组件): capture (camera, mic, keyboard, mouse, touch, pressure), storage (disk/flash/optical), network (LAN/4G/5G/Bluetooth/RF), computer/DSP (workstation/server; encode/analyse/fuse), output (LCD, speaker, vibration motor, haptics). Phone call: microphone -> digitize/encode/buffer -> network -> decode -> speaker.

Four problems (四类核心问题) (slide): (1) compact representation for storage/communication; (2) effective presentation of vast data; (3) system support for disparate data types; (4) combining streams to get task-essential information. *e.g.* a video call must compress, display, handle audio + video, and sync them.

Representation, data, and programming

Media as data: sound = 1-D time signal; image = 2-D pixel array; video = ordered image sequence. W x H = spatial sampling, **bit depth** (位深) = amplitude/color precision, FPS = temporal sampling, bitrate = coded bits per second.

Pixels: 1-bit = black/white; 8-bit grayscale = 256 levels; RGB 8 bits/channel = 24 bpp; RGBA = 32 bpp: standard alpha = opacity/coverage; the teacher described the 4th 8-bit channel as a mask (foreground / which window is on top). Colour hex #RRGGBB = 2 hex digits per channel (*e.g.* #FFCC00).

Size formulas: raw image bytes = W*H*bpp/8; raw video multiplies by FPS and seconds; PCM bps = sample rate*bit depth*channels; storage = bitrate*duration.

Teacher's 3 performance parameters: resolution (image quality), FPS (fluency; about 24-25 fps is the minimum to perceive motion; 24/25/30 by standard), kbps (compression efficiency, H.264/H.265).

Worked (class): 4x4 RGBA at 32 bpp = 16*32 = 512 bits = 64 B; 1080p RGBA raw frame = 1920*1080*4 B = 8,294,400 B about 8.3 MB, x25 for one second of video.

Synchronization (同步): streams/clocks must share one time reference.

Programming paradigms: procedural programming (面向过程) = steps/control flow (step 1 ... step 10; C); **OO programming** (面向对象) = class for an object's static properties, methods for events/dynamic behaviour between objects (C++/Java); **functional programming** (函数式) = state WHAT you want (*e.g.* "sort this"), not HOW, immutable data, few side effects (like AI coding). Python is multi-paradigm.

Binary and data structures: a circuit uses distinguishable physical states (open/closed) for 0/1 (Soviet Setun used 3-state logic). Data structures: list, array, table, set, map; humans do not store experience as vectors -> gap.

Sampling, quantization, Nyquist, and PCM

Sinusoid: x(t)=A cos(2pi f t + phi): A **amplitude** (振幅), f Hz, T=1/f, omega=2pi f, phi **phase** (相位) (projection of uniform circular motion); time shift dt -> phase shift 2pi f dt. A complex signal = sum A_k cos(2pi f_k t + phi_k).

Digitization: converting a continuous signal into a discrete set of its samples, in both time and signal value. Digitized values are much less susceptible to distortion than analog signals.

Sampling (采样): capture the analog signal at regular time intervals (sampling rate fs): x[n]=x(n/fs), Ts=1/fs. Error it introduces = **discretization error** (离散化误差) (limits the maximum frequency).

Quantization (量化): each sample is rounded to a fixed set of integer levels: n bits -> 2^n levels; step Delta about range/2^n, rounding error <= Delta/2 when no overload; error e=x-Q(x) = quantization error. Common depths 8/16/24/32 bits. （采样把时间变离散，量化把幅度变离散；采样率决定最高可表示频率，位深决定幅度精度。）

Trade-off: more samples improve temporal density; more bits reduce amplitude error; both increase data and cannot substitute for each other. 1-bit quantization destroys detail. Legacy telephone 8 kHz (bandwidth-limited) vs smartphone higher rates -> better call quality.

Class example: 5 s, 1 sample/s, 4 bits -> 5 samples*4 bits = 20 bits; bits = duration*fs*bits/sample*channels.

Nyquist theorem: if x(t) contains no frequencies higher than B Hz, it is completely determined by samples spaced 1/(2B) s apart. **Nyquist rate** (奈奎斯特率) = 2B; **Nyquist frequency** (奈奎斯特频率) = fs/2. Use fs > 2B (concert: drum low + flute high -> the flute's highest frequency sets B) plus an **anti-alias low-pass filter** (抗混叠低通滤波器) with margin.

Why not fs=2B: x=A cos(2pi B t+theta), Ts=1/(2B) -> x[n]=A(-1)^n cos(theta); phase lost, and theta=pi/2 gives all-zero samples.

Aliasing (混叠): different analog frequencies give the same samples: at 32 kHz, 18 kHz aliases to 14 kHz. 44.1/48/96 kHz -> Nyquist 22.05/24/48 kHz. Hearing 20 Hz-22 kHz -> record at >=44 kHz with filter margin; 48/96 kHz help digital filtering/ML, not audible range.

PCM calc: 44.1k*16*2=1.4112 Mb/s; 60 s = 10.584 MB about 10.09 MiB, excluding headers. Duration of N samples = N/fs.

Audio analysis, coding, and communication

ANC (主动降噪): a microphone measures noise, the circuit makes an opposite (180 deg) waveform, the speaker plays it: n + (-n) approximately 0 (destructive interference). Same idea in cars/buses: 3 ceiling mics + ECU + engine speed -> speakers play anti-noise. Limited by latency, phase/amplitude mismatch, position, changing sound.

Call noise reduction (通话降噪): a near-mouth main mic (speech + noise) plus reference mic(s) (noise) -> subtract/suppress; different from cancelling the ear sound field. More mics exploit spatial differences but cost size, power, **headroom** (峰值余量) (iPhone 4: 2 mics; 5s/6: 3; 6s: 4; Amazon Echo: 7). Constraints: distant speech, background noise, always-on power, overload, size.

EQ (均衡器): fine-tunes gain/attenuation of frequency bands across 20 Hz-20 kHz; presets (Bass Booster, Classical, Jazz, Rock) are gain curves; >200 bands in high-end systems; +/-6 to +/-24 dB. It changes **timbre** (音色); volume changes overall level. Cannot separate speech and noise in the same band; excessive boost distorts.

ASR (自动语音识别): waveform -> A/D -> denoise/normalize -> short segments (hundredths/thousandths of a second) -> match **phoneme** (音素) (~40 English phonemes) -> statistical/language model -> words. Accent, noise, homophones, mismatched conditions cause errors; ASR is not speaker recognition or translation.

Shazam: audio -> **spectrogram** (声谱图) -> salient peaks = **fingerprint** (音频指纹) of each catalogue song, indexed by song ID/time. A 10-s query gives the same landmarks; matching peaks vote for a common time **offset** (时间偏移) -> song + position, ranked by similarity score. Slide: robust to noise, room reverb, EQ, overlapping sounds (stable peaks survive), though not every distortion.

LPC (线性预测编码) vocoder: vocoder (声码器) = voice + coder; based on the human vocal model: air through the epiglottis vibrates vocal cords (voiced) and/or resonates in the mouth/nose cavity. Source (periodic pulses = voiced, noise = unvoiced) -> **vocal-tract filter** (声道滤波器). Send filter coefficients, energy, pitch, voiced/unvoiced; GSM voice is LPC-based; other use: speech synthesis / voice transformation. Models speech production, not a general music model.

Ear (for MPEG): eardrum vibration -> hydraulic pressure in the fluid-filled cochlea -> 20,000+ hair cells (high tones stimulate short hairs; base = high, apex = low frequency, like piano keys) -> the ear senses a spectrogram.

MPEG audio: lossy coding (有损编码) based on **psychoacoustics** (心理声学). Frequency masking (频率掩蔽): a loud sound in one band hides a softer sound in another band that would be audible without it. Temporal masking (时间掩蔽): after a loud sound stops, a quiet sound is inaudible for a short time (the ear turns its gain down and takes time to turn it up). Absolute threshold varies with frequency (most sensitive mid-kHz); a component below the raised masking curve is inaudible.

MP3: Layer I basic (DCC digital tapes), II advanced bit allocation, III adds hybrid filters; MP3 = MPEG Audio Layer III, not MPEG-3. 1152 samples (about 26 ms at 44.1 kHz) -> 32 bands -> masked frequencies via psychoacoustic model -> bit allocation/quantization (the lossy step) -> variable-length coding: **entropy/VLC coding** (熵编码/变长编码) removes redundancy without extra loss.

Media applications: imaging, video, vision, retrieval

HDR (高动态范围): fuses aligned complementary exposures: long exposure keeps shadows, short keeps highlights. "Dynamic" = brightness range, not motion. Needs registration; moving objects ghost.

Night Mode (夜景模式) / Deep Fusion: frames at various shutter speeds combined to recover shadow detail and avoid motion blur; Deep Fusion (iPhone 11 Pro, 2019) = 4 short + 1 long + 4 secondary = 9 frames, optimised for detail and low noise.

Super-resolution (超分辨率): shifted multi-frame samples improve spatial detail (also satellite images: "if one shot is not enough, take more").

Retinex (视网膜皮层增强): I=R*L; estimates/reduces illumination influence to enhance reflectance visibility/contrast; over-processing adds noise/halos.

Stabilization (防抖/稳像): stabilises shaky footage: track background feature points across frames -> estimate global camera motion -> compensate/smooth **trajectory** (轨迹); may crop edges; fails on moving foregrounds; does not remove within-exposure blur (drones, ships, phones do it in-chip).

Motion blur (运动模糊): motion accumulated during exposure, from global camera motion or fast objects; different parts of one object can have different velocities, so no single global motion model. FPS is not exposure time; higher ISO/gain does not collect more photons.

Deblurring (去模糊) pipeline: blurred video -> foreground mask -> segmentation/corners -> feature tracking (HBMA) -> motion estimate -> **blur kernel** (模糊核) -> **deconvolution** (反卷积)/restoration. Wrong kernels, noise and lost information prevent perfect recovery.

Vision tasks: detection (检测) = whether/where; **recognition** (识别) = identity; **tracking** (跟踪) = identity across frames; **reconstruction** (重建) = hidden/3-D structure; **retrieval** (检索) = ranks indexed material. A detection box is not identity.

Face detection (feature-based, Viola-Jones): Haar features (哈尔特征) = neighbouring light/dark rectangle-sum differences, computed efficiently; **AdaBoost** (自适应提升) combines weak classifiers into a strong one; a **cascade** (级联) rejects non-faces early; scan every location and scale.

THREE GENERATIONS of face recognition (teacher stressed): (1) feature/distance-based: eye-eye, eye-nose, nose-mouth distances as a "digital ID"; about 30-40% accuracy, fails for twins and changing expressions; (2) model-based: fit a general appearance model/mesh (<100 parameters; "Mission Impossible mask"), parameters = features, tolerates expression/angle; (3) deep learning: millions of weights learn features automatically, >80-90%, used at HK immigration and bank apps.

Head tracking (model-based): fit a q1-q8 spline head contour; a **tripwire** (虚拟绊线) counts people only when trajectories cross a virtual line, capturing the head there. False positives: faces on T-shirts, backpacks; occlusion and broken tracks cause errors.

3-D face reconstruction (三维人脸重建): recognition works well only on frontal faces, so convert non-frontal -> frontal: (1) feature detection: fit frontal/half-profile/profile AAMs, keep the minimum residual error; (2) 3-D modelling: personalise a generic **mesh** (网格) (6292 vertices, 6152 facets); **texture map** (纹理贴图) holds face texture, texture coordinates map 2-D texture onto the 3-D mesh; (3) **pose estimation** (姿态估计): rotation/location/scale; (4) texture rebuilding: fill/interpolate, postprocess, render frontal. Similarity scores rise, *e.g.* 0.33 -> 0.55, 0.00 -> 0.85. Hidden texture is inferred; reduces pose mismatch, not all errors.

Recognition limits: CMU: specially designed glasses frames fool recognition or impersonate someone; York: even ordinary glasses hurt HUMAN recognition of unfamiliar faces. Face ID adds structured IR/depth cues, not absolute anti-spoofing.

Gait (步态): "one of the most important biometrics"; identifies people at low resolution (few pixels); "KGB walk" (one arm still); Mission Impossible: face -> iris -> gait. Silhouette (轮廓) -> match a 16-pose **codebook** (码本) (dictionary idea, one-hot codeword per frame) -> Bag-of-Gait vector -> compare. Clothing, carried objects, speed, viewpoint matter.

Video retrieval: H.264/MPEG-4 stream -> motion vectors + background model -> moving-object segmentation -> tracking -> per-object features (size, colour histogram bins, edge-orientation bins 0/45/90/135 deg, trajectory) -> metadata index DB -> photo/sketch query -> ranked clips. Pyramid: video data -> metadata -> analytics -> insights. Teacher: the hard part is the index, not the search (library book ID; Google PageRank); similar colours do not prove the same object.

XR, DIKW, and city systems

VR (虚拟现实) / AR (增强现实): VR presents a virtual environment; AR overlays digital content on the real view; both need viewpoint tracking and low-latency rendering. Metaverse interview (Lex Fridman/Zuckerberg): static 3-D avatar + real-time expression/head parameters.

DIKW (数据-信息-知识-智慧) (slide): Data (acquire/tag/cleanse/correlate) -> Information (context; reports/dashboards) -> Knowledge (patterns; **predictive analytics** (预测性分析)) -> Wisdom (action; **prescriptive analytics** (决策建议型分析)). Uses: forecasting, recommendation, anomaly/fraud detection, maintenance. Example: recordings -> clean/ASR/labels -> grouped complaints -> hypothesized product issue -> repair decision; correlation alone is not causation.

Smart city (智慧城市): distributed sound-leak, gas, wind, bridge-vibration, video sensors -> network -> domain cleaning/fusion -> anomaly/prediction -> warning/control. A phone is also sensing -> analysis -> action. Teacher: understand the methodology/flow of each application, not implementation details.

Sensors, touch, errors, and waves

Sensor: a device that measures a physical quantity and converts it into a signal that a human or a machine can use. Pipeline: physical quantity -> transduction -> conditioning/amplification -> ADC -> analysis. Sound and light sensors matter most for multimedia. *e.g.* a thermometer turns temperature into a voltage then a number.

Sensor categories (传感器类别): (1) imitate human senses; (2) measure aspects of life; (3) measure physical facts about the environment.

Touch: skin = largest organ; tactile sensors translate pressure into signals (pressure, resistive, capacitive). **Resistive** (电阻式触控): pressing makes two conductive layers contact; robust to dust/water, needs a firm press (supermarket/7-Eleven self-checkout). Capacitive (电容式触控): finger changes capacitance; light touch, multi-touch; water/insulating gloves disturb it. 3-D Touch measures force. **Actuator** (执行器) gives feedback (vibration/Taptic Engine): input sensing differs from output feedback.

Sight: CCD/CMOS sensing elements (photovoltaic principle); more elements -> higher resolution. Brain (spot-the-difference): recognise objects -> **occipital** (枕叶); spatial relations -> occipital + **parietal** (顶叶); short-term memory -> **frontal** (额叶) + parietal; marking/movement -> frontal. **Binocular disparity** (双目视差)/stereograms give depth.

Hearing: audio sensors convert changes in air pressure into electrical signals (dynamic and condenser microphones).

Smell/taste: chemistry not well understood; no proper artificial nose/tongue yet. Other species sense temperature, gravity, magnetic fields, humidity, infrared.

Life sensors: weight (+ height/gender/age -> BMI = kg/m^2), body temperature, noise (loud environments, construction sites), heart rate, ECG (sinus rhythm), movement (+ heart rate -> calories, a model estimate). Care-on-Call watch for seniors -> call centre -> ambulance.

Environment sensors: Accelerometer (加速度计) (orientation, pedometer), **gyroscope** (陀螺仪) (rotation), GPS (location), radiation sensor, **altimeter** (高度计) (altitude). More reliable than human perception (spatial disorientation; you cannot "feel" 100 km/h); cars carry 50+ sensors.

Sensor limitations: (1) limited **dynamic range** (动态范围): only inputs within an interval (shadows vs highlights lost); fix: HDR or sensor arrays covering sub-ranges (0-120 + 120-240); (2) **offset** (偏置)/bias: nonzero output at zero input (y=kx+b), **calibrate** (校准); (3) **nonlinearity** (非线性): linear input change gives non-linear output; tune to be linear in an operating range; (4) **hysteresis** (迟滞): "sensor response deviation over time"; curve differs rising vs falling (5->10 gives 2->5, falling takes another

path, so output 3 maps to two inputs).
Sound: a travelling **longitudinal pressure wave** (纵向压力波), an oscillation of pressure through solid, liquid or gas; cannot travel through vacuum. *e.g.* a shout in air (no sound in space). Air 343 m/s (= Mach 1, 20 C), water 1482 m/s; lambda=v/f. Hearing: slides give 12 Hz-22 kHz and 20 Hz-22 kHz.
SPL: **Lp** (声压级)=20log10(p/pref)=10log10(p^2/pref^2) dB; p = **RMS** (均方根) pressure (sine: p_rms=p_peak/sqrt2); pref = 20 microPa air (threshold of hearing, a mosquito 3 m away), 1 microPa water. 0.2 Pa -> 80 dB; pressure ratio = 10^(dL/20); double pressure +6.02 dB, double power +3.01 dB. 0 dB is reference, not silence; dB is relative; do not normally add dB values. The ear is non-linear and less sensitive at low frequencies.
A-weighting (A计权): frequency weighting to match perceived loudness (ear less sensitive to low frequencies): add a table of values per octave/third-octave band, L_A(f)=L_p(f)+A(f) -> dBA. Same SPL at different frequencies need not sound equally loud.

Acoustics, microphones, recording, and loudspeakers

Echo (回声): a reflection of sound perceived some time after the direct sound (water well, mountains); humans cannot separate it if delay < 0.1 s, but machines may still "hear" multiple sources. Distance d=vt/2 (0.1 s -> 17.15 m).
Reverberation (混响): in an enclosed space numerous echoes build up and slowly decay as the room absorbs the sound; hurts content analysis even when inaudible.
Interference (干涉): superposition of two or more waves giving a new wave pattern: same f and A in phase -> 2A; 180 deg out of phase -> cancel (basis of ANC; imperfect match leaves residual).
Ear chain: pinna/canal -> eardrum/ossicles -> **cochlear** (耳蜗) hair cells -> auditory nerve.
Localisation: **ITD** (双耳时间差) = arrival-time difference; **ILD** (双耳声级差) = level difference (both -> left/right); **HRTF** (头部相关传递函数) = volume/quality changes from pinna reflections (front/back/up/down).
（声级差和时间差判断左右，耳廓滤波帮助辨别前后上下。）
Microphone: an acoustic sensor that converts sound into an electrical signal: a membrane exposed to sound pressure varies an electrical property (resistance, voltage, capacitance). *e.g.* a phone mic captures your voice for a call.
Dynamic microphone (动圈麦克风): diaphragm moves a coil in a permanent magnet's field -> induced voltage; no power supply needed. **Condenser** (电容麦克风): diaphragm/backplate distance changes C=epsilon*A/d; needs bias power; more sensitive.
Directionality / polar response (指向性): sensitivity to pressure waves arriving at different angles; polar pattern = locations giving the same output for a constant SPL source; varies with frequency. Omni (全指向): all angles, good in quiet studios, bad at rejecting noise; **cardioid** (心形): max rejection 180 deg; **supercardioid** (超心形): narrower front, rear lobe, max rejection 120 deg; **shotgun** (枪式): highly directional; **bidirectional** (双指向/8字形): front/rear.
Frequency response (频率响应): output ideally proportional to pressure; linear only for certain SPL and frequency ranges.
Gain (增益): analog output -> amplifier -> ADC. Too high: overdriven, **clips** (削波), distortion (unrepairable); too low: small values zero out after A/D. Keep headroom. Unity gain = x1 = 0 dB.
AES42 digital output: no analog interference problem; configurable polar pattern, low-cut (低切) 40/80/120 Hz (= **high-pass filter** (高通滤波器), removes rumble), gain unity to +63 dB, mute.
Early recording: sound pressure engraved on tinfoil around a rotating cylinder; a needle plays it back, mechanically amplified. An 18-note music box encodes notes, not arbitrary waveforms.
Loudspeaker: the most common sound-reproduction device; reverse of a microphone （麦克风：声→电；扬声器：电→声。）: current vibrates a diaphragm -> pressure waves in air/water; directional too.
Speaker specs: **rated power** (额定功率) = maximum input before damage (not loudness); **impedance** (阻抗) = electrical compatibility with amplifiers; crossover frequency (分频频率) = signal division by drivers; frequency range = linear response range.
Drivers: low frequencies are easier with a bigger diaphragm (larger mass vibrates slower), highs with small/light ones. Crossover: **low-pass** (低通) -> **woofer** (低音单元), high-pass -> **tweeter** (高音单元), band-pass -> **midrange** (中音单元).
Enclosure: **Cabinet** (箱体) prevents rear-diaphragm waves from cancelling front waves （振膜前后声波反相，直接相遇会削弱低音。）; **passive radiator** (无源辐射器) is moved by internal pressure, no voice coil.
Surround: 5.1 = front L/R, centre, surround L/R + **LFE** (低频效果声道); 6.1 adds rear centre; 7.1 adds rear L/R. Hard to deploy (space, wiring), so **virtual surround** (虚拟环绕声) (2.1, or a single strip = digital sound projection / **sound-projector array** (声音投射阵列)) recreates ILD, ITD, HRTF; depends on seat and room.

Speech and music

Speech ranges: normal speech 80 Hz-5 kHz; pitch male 120-160 Hz, women/children 220-330 Hz (children/women high, old men low); vowels up to 5 kHz; **sibilants** (咝音) (s, f) up to 20 kHz; 60 dBA normal speech at 60 cm, yelling 76 dBA male / 68 dBA female. Airflow excites **vocal folds** (声带); F0 = pitch; vocal-tract resonances form **formants** (共振峰).
Speech sounds (phonemes): Stops (塞音): blocked airflow (/p,t,k/); **fricatives** (擦音): partially blocked, turbulent (/f,v/); **approximants** (近音): slight turbulence (/w,r/); **aspirate** (送气音): exhaled breath (/h/); **vowels** (元音): full unimpeded airflow. Consonants + vowels = phonemes; vary by language.
Spectrogram signatures: time horizontal, frequency vertical, energy colour. **Voiced** (浊音): formant bands (yellow); unvoiced: no formants; stops: strong burst across all bands after a short silence; fricatives/aspirates: "large clouds" of smooth energy.
ASR limitations: people raise volume/change pitch/duration in noise, so quiet-trained models fail; phones sample at 8 kHz or below (Nyquist 4 kHz) so sibilants are lost; solution: spell with whole words ("Alpha, Beta, Charlie ... Sugar, Zero").
TTS (文字转语音) and dictation: text -> speech (Nuance demo; Hatsune Miku = synthesized singing); dictation -> NLP intent/entities -> action; perfect transcription does not ensure correct action.
Pitch (音高): a sound's perceived fundamental frequency (f0/F0).
Harmonic (谐波): k*f0; **partial** (分音): any component (overtones need not be integer multiples); **fundamental** (基音) = 1st harmonic; 1st **overtone** (泛音) = 2nd harmonic. 440/880/1320/1760 Hz = 1st-4th harmonics = fundamental/1st/2nd/3rd overtone.
Timbre: the quality that distinguishes different types of sound (*e.g.* piano vs violin on the same note); characterised by the frequency spectrum (the sum of distinct frequencies emitted for a note) and the time **envelope** (包络). Teacher's piano/violin question: same middle C (about 262 Hz) fundamental, different overtone mix -> different timbre.
ADSR (起音/衰减/持续/释音): Attack rise, Decay to Sustain, Sustain held level, Release after excitation stops.
Instruments: strings vibrate (excitation sets the harmonic mix); wind blows into a resonator; percussion is hit, shaken, rubbed, scraped.
Music-analysis goals: compress audio, detect instruments, manipulate/synthesise music.
Synthesis: physical modelling (harmonics + envelope/ADSR) or original samples (**wavetables** (波表)): generate wave shapes, several tones for fundamental + harmonics, vary volume over time. Storing F0 alone cannot recreate timbre (LEGO idea: build complex sound from simple sinusoids).

Light, photometry, imaging, and depth

Light: an **EM wave** (电磁波) with wavelength 400-780 nm; c=299,792,458 about 3*10^8 m/s, c=f*lambda. Teacher: amplitude = brightness/intensity (intensity proportional to E-field amplitude squared); frequency/wavelength = colour (white = mixture of all frequencies); phase = coherence.
Coherence: Monochromatic (单色光) = one frequency; **coherent** (相干) = stable relative phase (laser); light from normal sources is not in phase (non-coherent).
Polarization (偏振): describes the light wave's orientation; uncorrelated field orientations = unpolarized; filters pass one orientation. *e.g.* polarised sunglasses cut glare; 3-D cinema glasses. **Anaglyph** (红蓝立体) colour-filter stereo differs from polarization stereo.
Reflection (反射): incident ray, reflected ray and normal lie in one plane; angle of reflection = angle of incidence. **Specular** (镜面反射) = directional; **diffuse reflection** (漫反射) spreads.
Refraction (折射): the "bending" of light rays from one transparent material to another, because wavelength changes while frequency stays; n=c/v, n1 sin(theta1)=n2 sin(theta2); forms images in lenses/eyes. *e.g.* a straw looks bent in water. Mirrors and transparent objects complicate computer-vision detection.
Photometry (光度学): weights radiation by the **luminosity function** (光视效率函数) (average spectral sensitivity of human vision; **photopic** (明视觉) and scotopic curves differ); **radiometry** (辐射度学) uses watts, so electrical W is not lm.
Three units: candela cd = **luminous intensity** (发光强度), power emitted in a particular direction; lumen lm = **luminous flux** (光通量), the total light emitted; lux lx = **illuminance** (照度), luminous power per area. 1 lm = 1 cd*sr; isotropic 1 cd -> 4pi lm; 1 lx = 1 lm/m^2 (1000 lm on 1 m^2 = 1000 lx; on 10 m^2 = 100 lx).
（发光强度看方向，光通量看总输出，照度看单位面积接收多少光。）
Which unit when (teacher): buying a bulb/projector -> lm (total = sum over all directions); light toward one direction/receiver -> cd; desk/room lighting and camera low-light rating (minimum lux) -> lx. 450 lm = 40 W incandescent = 9-11 W CFL = 6-8 W LED; 100 W bulb about 120 cd. Office 320-500 lx, TV studio 1000 lx, desert sun up to 130,000 lx, full moon about 1 lx; eyes can detect a single photon.
Solid angle (立体角): radian cuts an arc equal to r; steradian cuts an area r^2; Omega = A/r^2, full sphere 4pi sr. Example: 50 mm*50 mm receiver at 2 m receives 0.1 W: A=.0025 m^2, Omega=.000625 sr, I=160 W/sr = **radiant intensity** (辐射强度), not cd. 1 cd = 540*10^12 Hz light at 1/683 W/sr.
Image recording: camera obscura ("dark chamber") / pinhole camera (针孔相机) projects but cannot store; first light-sensitive memory = paper with silver chloride; then film; now photosensitive electronics (light -> charge -> ADC).
Image formation: geometry (where a scene point lands on the image plane) + physics (brightness from illumination and surface properties).
Perspective: virtual upright image plane (虚拟正立像平面) x=fX/Z, y=fY/Z; size proportional to f/Z (larger f bigger, farther smaller); rear physical plane flips signs. Points on one ray share a projection, so one view loses depth.
（焦距增大则像变大，距离增大则像变小。） **Aperture** (光圈) controls light/**depth of field** (景深); long exposure risks blur.
Resolution / cameras: one photoelectric sensor = one pixel; MP=W*H/10^6; more pixels do not guarantee detail. Photo camera = still images (slide: up to 32 MP); video camera = sequence (FPS), compressed into one file (up to 16 MP). 720p=1280*720, 1080p=1920*1080, UHD 4K=3840*2160. Aspect ratio 4:3 (VGA 640*480, VCD/analog TV 720*576) -> 16:9 because two eyes give a wide field of view.
Stereo: separated viewpoints -> **disparity** (视差) d -> Z=fB/d (small disparity = far); phones carry 2-3 cameras like two eyes (close one eye, depth lost); multiple cameras give focal-length/FOV range and fusion, not just more megapixels.
ToF (飞行时间): active; emits modulated near-IR (850/940 nm), measures return delay/phase -> depth: Z=c*Delta t/2 (20 ns -> 3 m); multipath/ambient light limits.
Folded optics (折叠光路) / Leap Motion: prism folds a long lens path into a thin phone; Leap Motion uses IR cameras for hand pose (range/occlusion limits).

Displays, vision, color, and final distinctions

Subtractive (减色法): relies on reflected ambient light, absorbing parts of it; needs light. *e.g.* printed ink/toner on paper. CMYK; K not redundant (CMY black is muddy brown, 3x ink, wets paper, blurs text).
Additive (加色法): active light sources mixed together. *e.g.* a TV/monitor. RGB: zero = black, full mix = white. A red chair absorbs other colours and reflects red. Gamut (色域) = reproducible colour range; RGB->CMYK can lose colours.
Displays: CRT: electron gun beam excites phosphor; LCD: backlight modulated by **liquid crystal** (液晶)/polarizers/colour filters (TN twists; IPS in-plane, wider angle); OLED: pixels emit, no backlight, true black; AMOLED = active-matrix OLED. Early Kindle e-ink is reflective (no backlight).
Glasses-free 3-D: parallax barrier (视差屏障) sends different views to each eye; must sit in sweet spots or images blur. Gatebox "3D hologram" is a virtual-character display, not true holography.
Eye: iris (虹膜) controls light entering; lens focuses on the **retina** (视网膜) (cones and rods) -> chemical/nerve signals -> brain via optic nerve.
Photopic vision (bright): Cones (视锥细胞), 6-7 million, mainly in the **fovea** (中央凹), each with its own nerve end; colour + fine detail.
Scotopic vision (dim): rods (视杆细胞), 75-150 million, spread over the retina, several per nerve; general overall picture, no colour. Optic disc (视盘/盲点) has no receptors.
Resolving power: fovea about 1.5 mm x 1.5 mm x 150,000 cones/mm^2 = 337,500 (a medium CCD fits that in <= 5 mm x 5 mm); 1080p = 2,073,600 px (about 6x) but the fovea covers about 2 degrees. Eyes never stay still (Yarbus): the brain builds a higher-resolution image from snapshots; saccade (扫视) + **fixation** (注视); **foveated rendering** (注视点渲染) needs fast eye tracking (VR).
Subjective brightness (主观亮度): intensity as perceived by the visual system; a logarithmic function of incident light; context-dependent. *e.g.* the same grey looks brighter on a dark background. Brightness adaptation: about 10^10 overall, not simultaneously.
Colour spaces: CMYK for printing; RGB(A) for display (alpha = opacity); YUV for compression: **luma** (亮度) Y' and **chroma** (色度) U/V (eye less sensitive to colour -> chroma subsampling 4:2:0; B/W TV compatible; DVD component cables Y, Pb, Pr).
YUV formulas: Y'=.299R+.587G+.114B; U=-.14713R-.28886G+.436B; V=.615R-.51499G-.10001B; R=Y'+1.13983V; G=Y'-.39465U-.58060V; B=Y'+2.03211U. R=G=B -> U,V about 0. Y' is weighted, not mean(R,G,B); YCbCr conventions vary.
True Tone (原彩显示): senses ambient light and shifts the display's white so whites look the same under any lighting; not higher resolution or HDR.

Key comparisons (from the slides)

Multimedia vs monomedia (单媒体): monomedia (computer vision, audio processing) analyses one medium in isolation; multimedia follows how humans experience the world, correlating and combining partial information from multiple sources in the problem's context. *e.g.* COVID gate (thermal + camera) vs face detection alone.
Experiential vs current information environments: experiential = direct senses, same query/presentation space, user state/context, exploration; current = alphanumeric tables, formal queries/commands, separate result space, one medium, so humans do the logical work.
Human vs digital computer (人机差异): human = objects/events/concepts, perceptual and contextual, tolerates ambiguity, learns by experience; computer = bits/bytes in lists/arrays/tables, exact fast maths/logic, no meaning without models -> semantic gap.
Experience vs information vs knowledge: experience = direct observation/participation; information = abstract communication of it (the act of experiencing is lost); knowledge = accumulated experiences.
Objects vs events: objects = static components; events = changes in relationships among objects over time. *e.g.* person vs person entering a restricted zone.
Content vs context vs metadata: content = acquired data (pixels); context = conditions of acquisition; metadata = data about data giving context (Exif time/GPS).
Procedural vs OO vs functional: steps (how, step by step) vs classes = objects + methods = events vs declare what you want (not how).
Sampling vs quantization: sampling discretises time (fs; error = discretization; sets max frequency fs/2); quantization discretises amplitude (n bits; error = quantization noise; sets precision). Neither replaces the other.
Nyquist rate vs Nyquist frequency: Nyquist rate = 2B, the minimum sampling rate for highest frequency B; Nyquist frequency = fs/2, the highest frequency a given fs can represent.
Resistive vs capacitive touch: both layered sensors giving x-y. Resistive = pressure makes two conductive layers touch; any object/glove, dust/water-proof, firm press, single touch. Capacitive = finger changes capacitance; light multi-touch, no ordinary gloves, water disturbs.
Dynamic vs condenser mic: both diaphragm -> analog voltage. Dynamic = coil in magnet (induction), no power, robust, high SPL. Condenser = capacitor plates (C=epsilon*A/d), needs power, more sensitive, studio.
Omni vs cardioid vs supercardioid vs shotgun: all angles (quiet studio, conference table) vs front, null 180 deg vs narrower, rear lobe, null 120 deg vs highly directional.
Echo vs reverberation vs interference: one delayed reflection (merged if < 0.1 s) vs many decaying reflections in a room vs superposition (in phase 2A, anti-phase cancel = ANC).
ITD vs ILD vs HRTF: time difference vs level difference (both left/right) vs pinna filtering (front/back/up/down).
Frequency vs temporal masking: loud sound hides a softer one in a nearby band (simultaneous) vs a quiet sound just after a loud one is hidden (ear gain recovering).
LPC vs MPEG audio: LPC models speech production (source-filter, parameters, GSM, speech only); MPEG models hearing (psychoacoustic masking, bit allocation, any audio). "How was it produced?" vs "which errors are hard to hear?".
Stabilization vs deblurring: frame-to-frame camera shake (track background, compensate) vs blur within one exposure (blur kernel, deconvolution).
Detection vs recognition vs tracking: is there a face and where vs whose face vs same identity across frames.
Feature-based vs model-based: face detection = Haar features + AdaBoost; head tracking / 3-D reconstruction = fit a general model (spline, AAM, mesh); model-based tolerates expression and pose.
VR vs AR: fully virtual environment vs digital content overlaid on the real view.
Pitch vs timbre: perceived fundamental F0 vs spectrum + envelope quality. Harmonic = k*f0; overtone = harmonic above the fundamental (1st overtone = 2nd harmonic); partial = any component.
Photometry vs radiometry: weighted by human sensitivity (cd, lm, lx) vs physical power (W, W/sr).
Candela vs lumen vs lux: one direction (spotlight) vs total output (bulb, projector) vs per area received (desk, camera low-light rating). 1 lm = 1 cd*sr; 1 lx = 1 lm/m^2.
Reflection vs refraction: bounces back (angle in = angle out) vs bends into a new medium (wavelength changes, frequency fixed). Specular (mirror, directional) vs diffuse (scattered).
Monochromatic vs coherent vs polarized: one frequency vs stable relative phase (laser) vs fixed field orientation.
CCD vs CMOS (slide table): pixel transfer shift register vs direct pixel address; charge->voltage after readout vs in-pixel; exposure single frame vs frame/line-sequential; sensitivity higher vs lower; control circuit off-sensor vs integrated. CCD better low light; megapixel phones use CMOS.
Stereo vs ToF: passive, two views, Z=fB/d vs active near-IR, Z=c*Delta t/2.
Photo vs video camera: still images (up to 32 MP) vs image sequence at FPS, compressed into one file (up to 16 MP).
Additive vs subtractive: emit and mix light (RGB, screens, zero = black) vs absorb ambient light (CMYK ink, needs light, white paper).
RGB vs CMYK vs YUV: display vs printing vs compression (luma + chroma, chroma subsampled).
LCD vs OLED: backlight + liquid crystal + polarisers (grey black, thicker) vs self-emitting pixels (true black, thin, flexible).
Photopic vs scotopic: cones 6-7M, fovea, own nerve each, colour + detail, bright light vs rods 75-150M, periphery, shared nerves, dim light, no colour.
Luminance vs luma vs subjective brightness: physical photometric light vs weighted Y' signal (0.299R+0.587G+0.114B) vs perceived, logarithmic, context-dependent.

Past-paper answer cues (finals 2015-2025: Q1-Q2 = L1-L5)

Most asked: experiential environment (6x), sampling + quantization effects (5x), human vs computer (4x), Neisser/perception diagram, DIKW, mic digitization.
Digitize audio with a mic (model answer): sound pressure -> diaphragm -> analog voltage (dynamic/condenser) -> preamp gain (avoid clipping, keep headroom) -> anti-alias low-pass -> sample at fs > 2B (Nyquist; 44.1 kHz for 20-22 kHz hearing) -> quantize to n bits (quantization noise = rounding error, SNR about 6.02n dB) -> encode PCM -> store/transmit.
Sampling-rate choice: signal bandwidth (Nyquist), human perceptual limits, storage/bandwidth cost, processing/ML needs, anti-alias filter margin. Standards: CD 44.1 kHz (20 kHz audio), telephone 8 kHz (about 3.4-4 kHz speech), PAL 25 / NTSC 30 fps.
Effects per medium (采样/量化对质量的影响): AUDIO low fs -> lost highs + aliasing, few bits -> hiss, low dynamic range. IMAGE low spatial sampling -> pixelation, jagged edges, moire; few bits -> banding/false contours. VIDEO low fps -> jerky motion, wagon-wheel aliasing; each frame as image.
Quantization = "the other process": A-law/mu-law = non-uniform (perceptual) quantization, 8 bits at 8 kHz = 64 kb/s, finer steps for quiet sounds (Weber-Fechner: perceived change proportional to dI/I).
SPL logarithmic: pressures span about 10^7 (20 microPa to 200 Pa) and hearing responds roughly logarithmically.
Sinusoid for sound vs light: SOUND A = pressure (Pa, dB SPL) -> loudness, f (Hz) -> pitch, phi -> interference; LIGHT A = E-field -> intensity (cd, lm, lx), f or lambda (nm) -> colour, phi -> coherence.
Speech in frequency domain: phonemes show as formant patterns; the waveform changes with loudness/phase; the cochlea analyses frequency.
Conference-table mic: omnidirectional (participants all around). Polar pattern measured by rotating the mic before a constant-SPL source, per angle, at several frequencies.
Nit (cd/m^2): suits displays: luminance per area is independent of screen size and viewing distance. Phone cameras: lenses close together to minimise parallax; ultra-wide/wide/tele = zoom range + fusion + depth (bokeh), HDR/night.
Calcs seen: CD 44.1k*16*2 = 1.4112 Mb/s; 5 min CD 52.92 MB vs A-law 2.4 MB -> about 22:1; FHD RGB24 PAL 1920*1080*24*25 = 1.244 Gb/s; fovea 337,500 vs 1080p 2,073,600.